

ANALYSTLAB AFRICA

Experience Lab Internship Programme



HealthConnect Clinic

ML Workflow Design

By

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Project: HealthConnect Clinic Experience Lab – Improving Patient Appointment Attendance and Healthcare Support Using Data and AI

Track: Data Science

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Week 4 Submission – Supporting Document

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1. Introduction

This document describes the proposed end-to-end operational workflow for the HealthConnect no-show prediction solution: the sequence of steps that would occur, in practice, from the moment a patient books an appointment to the point their attendance outcome is recorded and used to improve the model. It complements the ML System Architecture Design document, which describes the system's components, by focusing instead on the process flow between them.

As with the architecture document, this is a Week 4 planning artefact. No part of this workflow has been implemented; it is intended to guide Week 5 development and to give the Machine Learning Engineering and Project Management tracks a shared, concrete picture of how the prediction solution is expected to operate.

2. Workflow Overview

The workflow has two related cycles. The first is a per-appointment cycle: each time an appointment is booked, the system extracts the features available at that moment, scores the appointment for no-show risk, and either follows the standard reminder process or flags the appointment for staff outreach. The second is a longer retraining cycle: once an appointment's actual outcome is known, it is logged and periodically used to refresh the underlying model, so that the system continues to reflect current patient behaviour.

3. Step-by-Step Workflow Description

1. **New appointment booked:** a patient books an appointment through the clinic's booking channel or reception, generating a new record with booking-time information (patient details, appointment type, date, time, lead time, and reminder preference).
2. **Extract booking-time features:** the system pulls together the features that are legitimately known at this point, including the patient's own history of previous appointments and no-shows and the calculated booking lead time.
3. **Run the trained no-show risk model:** the prediction service scores the appointment and returns a no-show risk score.
4. **Risk threshold decision:** the risk score is compared against an agreed threshold, to be calibrated in Week 5 once real model output is available.
5. **Standard path (risk below threshold):** the appointment proceeds through HealthConnect's normal reminder workflow (SMS, WhatsApp or email, per the Clinic Knowledge Base).
6. **High-risk path (risk at or above threshold):** the appointment is flagged on the clinic staff risk dashboard, and staff prioritise outreach, such as a confirmation call or an offer to reschedule.
7. **Appointment date arrives:** both paths converge on the scheduled appointment date.
8. **Outcome recorded:** clinic staff record the actual outcome (Attended, No-Show, or Cancelled), consistent with the `appointment_outcome` field in the current dataset.
9. **Outcome logged for retraining:** the outcome is added to the training data store, feeding the periodic retraining cycle described in Section 6.

4. Workflow Diagram

HealthConnect No-Show Prediction — Operational Workflow

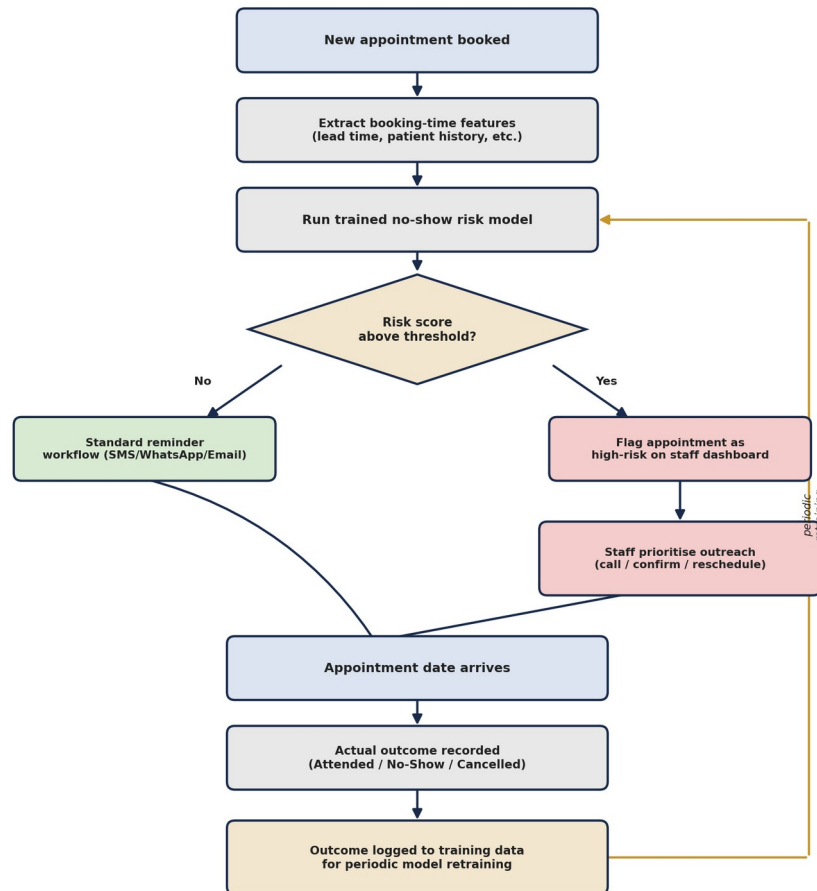


Figure 1. Proposed operational workflow for the HealthConnect no-show prediction solution.

5. Exception Handling and Edge Cases

- **Missing or incomplete booking-time features:** if distance_to_clinic_km or another feature is missing at scoring time, the system should fall back to a documented default (for example, a cohort median) rather than fail to produce a score.
- **New patients with no appointment history:** previous_appointments and previous_no_shows will be zero by definition; the model should be tested specifically on this group in Week 5, since it may behave differently from returning patients.
- **Cancelled appointments:** as set out in the Problem Definition Document, a cancellation is treated as a distinct, proactive outcome. Cancelled appointments should not be scored for no-show risk once the cancellation is recorded, and should not be treated as a missed prediction opportunity.
- **Same-day or very short lead-time bookings:** given the strong observed relationship between lead time and no-show risk, very short lead times should be handled carefully; the model should not be allowed to assign unrealistically low risk purely because lead time is short if other risk factors are present.

- **Reminder channel unavailable:** where reminder_sent is 'No' (and reminder_channel is therefore not applicable), the standard-path workflow should default to a documented fallback communication method rather than skipping reminders altogether.

6. Feedback and Retraining Loop

Recorded outcomes are fed back into the training dataset to support periodic retraining, allowing the model to adapt as patient behaviour or clinic operations change over time. This closes the loop shown in gold in Figure 1 and is a deliberate design choice: without a structured feedback and retraining process, a deployed model's performance can silently degrade as the patterns it originally learned drift out of date, a well-documented risk in operational machine learning systems (Sculley et al., 2015). The retraining cadence itself (for example, weekly or monthly) is a decision for the Machine Learning Engineering track and has not been fixed at this planning stage.

References

- AnalystLab Africa. (2026). *AnalystLab Africa Experience Lab – Week 4: HealthConnect project kickoff & problem understanding*. Assignment brief.
- AnalystLab Africa. (2026). *HealthConnect Clinic: ML system architecture design*. Week 4 submission.
- AnalystLab Africa. (2026). *HealthConnect Clinic: Machine learning problem definition document*. Week 4 submission.
- Sculley, D., Holt, G., Golovin, D., Davydov, E., Phillips, T., Ebner, D., Chaudhary, V., Young, M., Crespo, J.-F., & Dennison, D. (2015). *Hidden technical debt in machine learning systems*. In *Advances in Neural Information Processing Systems 28* (NeurIPS 2015) (pp. 2503–2511). <https://papers.nips.cc/paper/5656-hidden-technical-debt-in-machine-learning-systems>